Watson, ‘Brazil Election: “We’ll Vote for Bolsonaro Because He Is God”’, BBC, 28 Sept. 2022, www.bbc.com/news/world-latin-america-62929581.

28. Oliver Hahl, Minjae Kim and Ezra W. Zuckerman Sivan, ‘The Authentic Appeal of the Lying Demagogue: Proclaiming the Deeper Truth About Political Illegitimacy’, *American Sociological Review* 83, no. 1 (2018): 1–33.

CHAPTER 1: WHAT IS INFORMATION?

1. See, for example, the works of Nick Bostrom and David Chalmers on the simulation hypothesis. If the simulation hypothesis is true, then we have no idea what the universe is ultimately made of, but everything we see in our simulated world is made of bits of information. Nick Bostrom, ‘Are We Living in a Computer Simulation?’, *Philosophical Quarterly* 53, no. 211 (2003): 243–55, www.jstor.org/stable/3542867; David J. Chalmers, *Reality+: Virtual Worlds and the Problems of Philosophy* (New York: W. W. Norton, 2022). See also Archibald Wheeler’s influential notion of ‘it from bit’: John Archibald Wheeler, ‘Information, Physics, Quantum: The Search for Links’, *Proceedings III International Symposium on Foundations of Quantum Mechanics* (Tokyo, 1989), 354–68; Paul Davies and Niels Henrik Gregersen, eds, *Information and the Nature of Reality: From Physics to Metaphysics* (Cambridge: Cambridge University Press, 2014); Erik Verlinde, ‘On the Origin of Gravity and the Laws of Newton’, *Journal of High Energy Physics* 4 (2011): 1–27. It should be emphasised that while the ‘it from bit’ position is becoming more acceptable in physics, most physicists still doubt or reject it and believe that matter and energy are the fundamental building blocks of nature, while information is a derived phenomenon.

2. My understanding of information has been deeply influenced by Cesar Hidalgo, *Why Information Grows* (New York: Basic Books, 2015). For alternative views and discussions, see Artemy Kolchinsky and David H. Wolpert, ‘Semantic Information, Autonomous Agency, and Non-equilibrium Statistical Physics’, *Interface Focus* 8, no. 6 (2018), article 20180041; Peter Godfrey-